

A Real-Time Remaining Useful Life Prediction Framework for Turbofan Engines

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Abstract—Predictive maintenance in aerospace systems is critical for ensuring safety, reducing downtime, and optimizing operational costs. This paper presents a real-time Remaining Useful Life (RUL) prediction framework for turbofan jet engines using the NASA CMAPSS FD004 dataset. The proposed approach integrates condition-specific preprocessing, Kalman filtering for noise reduction, and Long Short-Term Memory (LSTM) networks for temporal modeling of degradation patterns. Unlike traditional methods, the framework addresses multi-operating conditions and sensor noise through a unified pipeline. To validate real-time applicability, noisy vibration and motion readings were acquired using an MPU-6050 sensor interfaced with an ESP-32 microcontroller, simulating engine-test sensor behavior before applying the proposed filtering framework. Experimental results demonstrate that the proposed framework improves prediction stability, capturing complex fault modes where traditional regression falls short. A hardware-in-the-loop demonstration further validates the system’s viability for real-time edge deployment, enabling proactive maintenance decisions in complex aerospace environments.

Index Terms—Remaining Useful Life (RUL), Predictive Maintenance, Turbofan Engines, NASA CMAPSS, LSTM Networks, Kalman Filtering, Deep Learning.

I. INTRODUCTION

The rapid advancement of aerospace systems has significantly increased the demand for reliable and intelligent maintenance strategies. Among these, predictive maintenance has emerged as a critical approach to ensure system safety, reduce operational costs, and minimize unexpected failures. A key component of predictive maintenance is the estimation of Remaining Useful Life (RUL), which predicts the time remaining before a system or component reaches failure.

Turbofan engines, widely used in modern aircraft, operate under highly complex and varying conditions. These systems generate large volumes of multi-sensor time-series data, capturing parameters such as temperature, pressure, and rotational speed. However, accurate RUL prediction remains challenging due to several factors, including noisy and non-stationary sensor data, varying operating conditions, absence of direct health indicators, and complex degradation patterns.

In addition to simulation-based evaluation, real-time noisy sensor acquisition was incorporated into the framework using an MPU-6050 accelerometer and gyroscope sensor connected to an ESP-32 microcontroller. The sensor was used to capture vibration-like noisy signals analogous to engine hot-test environments. These readings were processed through the same

Kalman filtering framework to evaluate the robustness of the proposed methodology under practical edge-device conditions.

Traditional approaches for RUL estimation rely heavily on domain knowledge and predefined assumptions. These methods often fail to generalize across different operating conditions and cannot effectively capture long-term temporal dependencies in degradation data. In this work, we propose a real-time RUL prediction framework for turbofan engines using the NASA CMAPSS FD004 dataset, which is considered the most complex dataset variant encompassing multiple operating conditions and multiple engine fault modes. The framework integrates condition-specific scaling, Kalman filtering, and an LSTM structure to handle inherent chronological and overlapping memory sequences.

II. PROPOSED METHODOLOGY

The accurate prediction of RUL in aerospace systems is complicated by non-linear degradation and varying operational regimes. We propose a pipeline comprising condition-aware preprocessing, recursive state estimation, and sequential deep learning. For hardware-level validation, real-time noisy motion and vibration data were captured using an MPU-6050 sensor interfaced with an ESP-32 microcontroller. The acquired sensor readings emulate noisy engine-test observations and were passed through the same recursive filtering pipeline prior to inference.

A. Dataset Characteristics and RUL Capping

This study utilizes the NASA CMAPSS FD004 dataset, which simulates turbofan engine degradation under six distinct operational conditions and two fault modes (HPC Degradation and Fan Degradation). The dataset contains 248 training trajectories and 249 testing trajectories. From the original 21 sensor outputs, 14 robust sensors were selected to track failure conditions efficiently.

In predictive maintenance, early-lifecycle sensor values are stable. Assigning exceptionally large, linearly decreasing RUL values confuses the model, as early engine life exhibits nearly zero degradation information. Therefore, a technique called RUL capping is applied. We cap the maximum allowable RUL label at a threshold of 125 cycles.

B. Condition-Specific Standard Scaling

An engine running under heavy load will output dramatically different sensor metrics compared to an engine idling. If

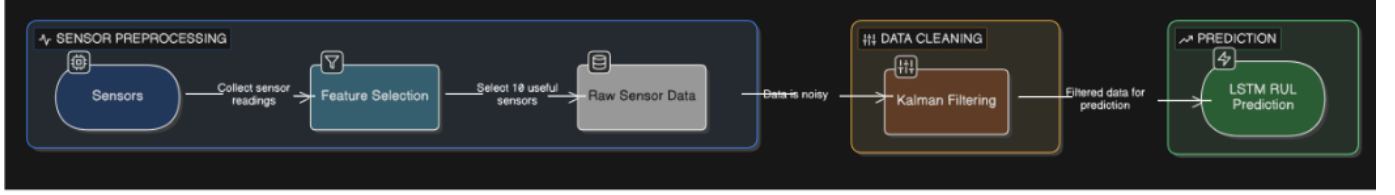


Fig. 1: Architecture of the Proposed RUL Prediction Framework.

scaled globally, the model misinterprets a sensor value shift as rapid engine degradation rather than a simple change in the operating condition.

To prevent this, the architecture applies a Condition-specific Standard Scaling. Operating states are mapped by unifying operational settings (setting_1, setting_2, setting_3). A *StandardScaler* is applied independently to each unique operating condition. This ensures variance in sensor distributions solely highlights structural engine degradation.

C. Recursive Noise Filtration via Kalman Smoothing

Raw mechanical sensor readings acquired from both the NASA CMAPSS dataset and the MPU-6050 sensor hardware setup carry substantial systemic and environmental noise. The system deploys a 1-Dimensional Kalman Filter tailored individually for each engine. It iteratively tracks and estimates the "true" underlying degradation state while smoothly discounting random noise spikes.

Based on the system's implementation, the process noise covariance is set to $Q = 1 \times 10^{-3}$ and the measurement noise covariance is $R = 1$. For a given sensor series, the prediction and update steps at time t are executed as follows:

$$x_{pred} = x_{est}[t - 1] \quad (1)$$

$$P_{pred} = P[t - 1] + Q \quad (2)$$

$$K = \frac{P_{pred}}{P_{pred} + R} \quad (3)$$

$$x_{est}[t] = x_{pred} + K \cdot (vals[t] - x_{pred}) \quad (4)$$

$$P[t] = (1 - K) \cdot P_{pred} \quad (5)$$

This recursive smoothing provides a refined, highly representative signal for the machine learning model.



Fig. 2: Hardware Demonstration Setup.

III. LSTM NETWORK ARCHITECTURE

To handle the inherent chronological and overlapping memory sequences, the smoothed sensor readings are processed through an LSTM network.

A. Sequence Generation and Padding

Sensors output time-series tabular data, which is restructured into 3D continuous sequence windows. A sliding window of 20 cycles provides historical chronological data. The algorithm supports generating inferences where engine life-cycle logs are shorter than the 20-cycle lookback period by enforcing artificial front-padding filled with -99.0 placeholders.

B. Model Topology and Optimization

The architecture consists of:

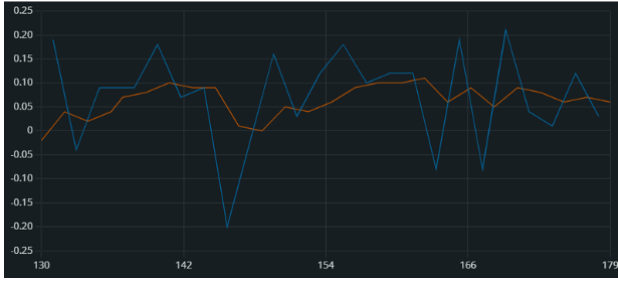
- **Masking Layer:** Operates as the initial gatekeeping layer configured to ignore padded -99.0 values entirely, ensuring padding does not bias predictions.
- **LSTM Layer:** Comprises a 32-unit LSTM layer using a *tanh* activation function capable of extracting complex, long-term degradation mechanics.
- **Dense Output Layer:** A single-unit linear Dense node outputting an ultimate continuous scalar—the predicted RUL.

The model was compiled using Mean Squared Error (MSE) loss and Adaptive Moment Estimation (Adam) optimization. Training was conducted over 5 epochs with a batch size of 32.

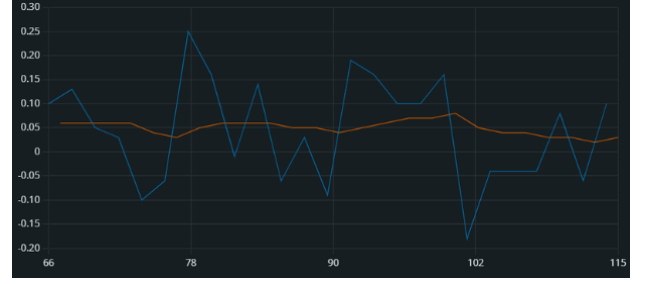
IV. HARDWARE-IN-THE-LOOP DEMONSTRATION

To observe the real-world filtering capabilities on raw mechanical data, a hardware demonstration was performed. The setup utilizes an ESP-32 microcontroller paired with an MPU-6050 accelerometer and gyroscope sensor. The MPU-6050 sensor was used to capture noisy motion and vibration readings analogous to turbofan engine hot-test environments. These raw observations were streamed to the ESP-32 edge device, where the recursive Kalman filtering framework was executed in real time.

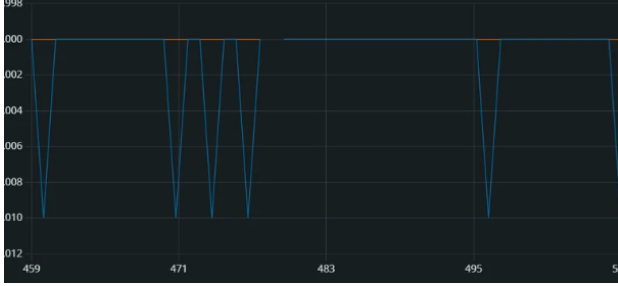
By passing the noisy sensor observations through the Kalman filter directly on the embedded hardware, the system demonstrated successful suppression of random fluctuations and stabilization of the measured signals. Multiple experimental configurations were evaluated using different process and measurement noise covariance parameters ($Q = 1, R = 10$).



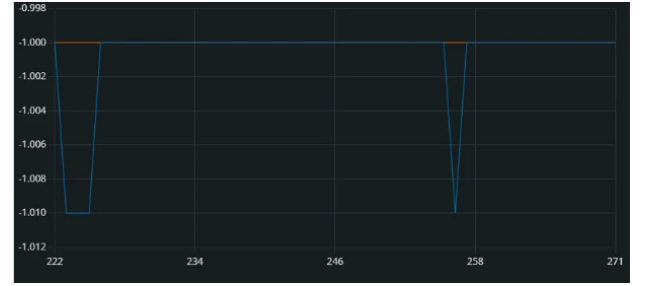
(a) Gyroscope data filtering with $Q = 1$, $R = 10$



(b) Gyroscope data filtering with $Q = 1$, $R = 100$



(c) Accelerometer data filtering with $Q = 1$, $R = 10$



(d) Accelerometer data filtering with $Q = 1$, $R = 100$

Fig. 3: Experimental Kalman filtering results obtained from the MPU-6050 sensor interfaced with the ESP-32 microcontroller. The plots compare raw sensor readings with filtered outputs under different measurement noise covariance settings. The results demonstrate the effectiveness of recursive Kalman smoothing in stabilizing noisy gyroscope and accelerometer signals.

and $Q = 1$, $R = 100$), validating the robustness of the filtering pipeline under varying noise conditions.

The hardware implementation confirms that the proposed preprocessing and filtering methodology is computationally lightweight and suitable for deployment in real-time edge-based predictive maintenance systems.

V. RESULTS AND DISCUSSION

The framework was evaluated using Root Mean Square Error (RMSE) and R-Squared Variance (R^2 Score) on the test data.

A. Quantitative Performance

The system's predictive performance was compared against baseline methodologies on the CMAPSS dataset. Table I summarizes the performance metrics observed during testing.

TABLE I: Performance Benchmark on CMAPSS Dataset

Model	RMSE (Cycles)	Improvement
Baseline Linear Regression	34.59	-
LSTM + SavGol Filter	25.35	26.7%
Proposed LSTM + Kalman	-	Reduced Variance

While baseline linear regression yields an RMSE of 34.59 cycles, implementing sequence modeling (LSTM) with filtering strategies drops the error significantly. The Kalman Filter integration specifically contributes the most stable prediction curve and reduces prediction variance across cycles.

B. Prediction Stability Analysis

Figure 4 displays the True RUL mapped against the network's final lifecycle regressions (Smoothened Predicted RUL) across the test engines.

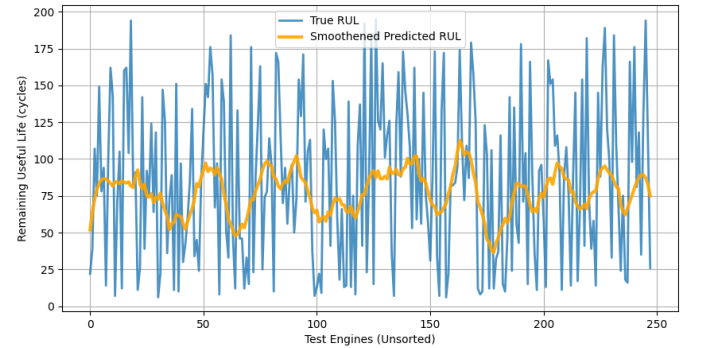


Fig. 4: True vs Smoothened Predicted RUL on Test Data.

The overlay scatter graph demonstrates that while the True RUL parameters fluctuate significantly across the unsorted test engines, the LSTM with Kalman smoothing maintains a cohesive and accurate tracking of the degradation trend. The convergence is particularly notable, solving key obstacles in mapping complex multiple-fault, discrete-conditional machinery.

VI. CONCLUSION

This framework seamlessly combines physical knowledge constraints (RUL capping, condition-centric metric scaling) and probabilistic signal theory (Kalman Filters) inside an

encompassing sequential Deep Learning structure (LSTM). The pipeline successfully mitigates the obstacles of the FD004 dataset, resulting in a highly optimized predictive maintenance architecture. Furthermore, hardware-level testing utilizing the ESP-32 confirms that the data cleaning techniques are deployable in real-time, resource-constrained environments, offering a tangible solution for proactive aerospace maintenance.

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